

Learning Resource On Speech and Video Processing

Module-1: Speech Processing Concepts

Chapter-4: Frequency Analysis of Speech Signal

Prepared By:

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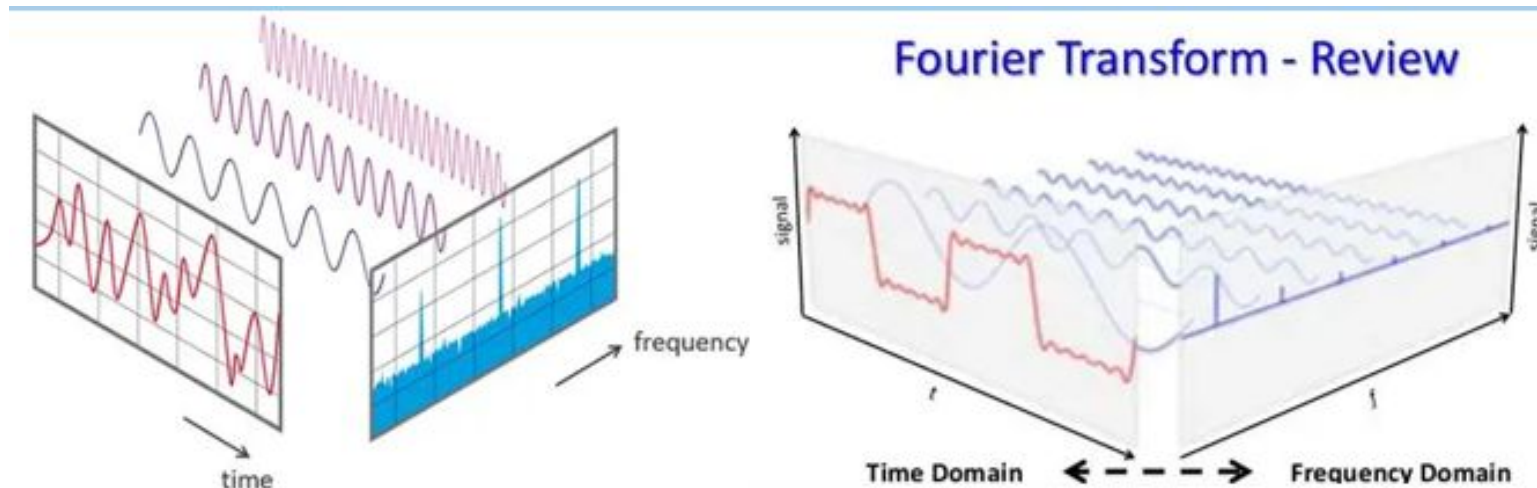
Chapter Outcomes

- After completing this module, students will be able to:
 - Explain the role of DFT and FFT in speech signal analysis.
 - Interpret the frequency spectrum of speech signals using DFT/FFT.
 - Explain the concept and importance of spectral estimation in speech processing.
 - Describe the working principle of linear filter banks for speech analysis.
 - Relate frequency-domain analysis techniques to real-world speech processing applications.

Fourier Transform

- *Why Frequency domain Representations instead of time-domain representations of signals and systems?*
- Speech signals are **non-stationary**. Time-domain provides **amplitude, duration** but hides **pitch, formants, harmonics**.
- While the time domain shows *when a signal changes*, the frequency domain shows *what frequencies* make up the signal.
- Frequency domain representation can be obtained by applying **Fourier Transform or Fast Fourier Transform** on the time-domain representation of the speech signal.
- **Fourier Transform** is used for *frequency-domain representations of a signal*. The aim is to understand and analyse the **underlying frequency** components causing the signal pattern.

- **Fourier transform** decomposes a signal in terms of **sinusoidal** (or complex exponential) components with different **frequencies, amplitudes, and phases**.
- The Fourier transform is one of several mathematical tools that is useful in the analysis and design of linear time- invariant (LTI) systems.
- Used for **spectral analysis** of speech.



Exponential Representation of Signal

- Euler's formula:

$e^{j\theta} = \cos(\theta) + j\sin(\theta) \rightarrow e^{j\theta}$ represents a rotating vector.

- A **sinusoidal signal** is represented by

$$s[n] = A \sin(\omega n + \phi) \text{ or } s[n] = A \cos(\omega n + \phi)$$

where, A = amplitude, ω = digital angular frequency($rad/sample$), n = sample index.

- Euler's formula can express sinusoids as complex exponentials

$$\cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

- Consider the signal: $3\cos(3\pi t) + 4\cos(2\pi t) + 5\cos(4\pi t)$; where $A=\{3,4,5\}$ and Angular frequency = $(3\pi, 2\pi, 4\pi)$ rad/s.
- Each sinusoid has two complex exponentials.
- As $\omega=2\pi f$, Here complex exponentials are : $\frac{3}{2}, 2, \frac{4}{2}$ corresponding to frequencies $f= \pm 1.5, \pm 1, \pm 2$ Hz.
- As the FT and DFT can't be applied directly on these cosines, the signal is represented in the form of complex exponentials using Euler's formula.
- So the signal can be represented as:

$$x(t) = \frac{3}{2}e^{j3\pi t} + \frac{3}{2}e^{-j3\pi t} + 2e^{j2\pi t} + 2e^{-j2\pi t} + \frac{5}{2}e^{j4\pi t} + \frac{5}{2}e^{-j4\pi t}$$

Fourier Transform

- **Fourier transform converts waveform → frequency spectrum**
- Fourier transform converts a signal from **time domain to frequency domain**, revealing the frequencies present in a signal and their corresponding amplitudes.
- For a **continuous-time signal $x(t)$** , the Fourier Transform is defined as

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

- *where, $x(t)$ = time-domain signal,
 $X(f)$ = frequency-domain representation,
 f = frequency (Hz)*

Example

- Let's consider the following signal: $x(t) = \cos(2\pi f_0 t)$

$$x(t) = \frac{1}{2} e^{j2\pi f_0 t} + \frac{1}{2} e^{-j2\pi f_0 t}$$

- The Fourier Transform of a complex exponential is

$$\mathcal{F}\{e^{j2\pi f_0 t}\} = \delta(f - f_0)$$

- Similarly, $\mathcal{F}\{e^{-j2\pi f_0 t}\} = \delta(f + f_0)$

- Its Fourier Transform will consist of two impulses:

$$X(f) = \mathbf{Ft}\left\{\frac{1}{2} e^{j2\pi f_0 t} + \frac{1}{2} e^{-j2\pi f_0 t}\right\} = \frac{1}{2} \delta(f - f_0) + \frac{1}{2} \delta(f + f_0)$$

- Here, $\delta(f - f_0)$, known as **dirac delta function** indicates that there is energy only at the frequency f_0 and there is no energy at any other frequency

- A real cosine contains two frequency components:
 - one at $+f_0$
 - one at $-f_0$
- Can we have negative frequency???
 - A negative frequency does not mean the signal oscillates backward in time. Instead, it comes from the complex exponential representation: $e^{j2\pi f_0 t}$ represents a counterclockwise rotating phasor and $e^{-j2\pi f_0 t}$ represents a clockwise rotating phasor.
- A real cosine requires both rotations. Therefore, its spectrum is symmetric about zero frequency.
- Suppose someone sustains the vowel "aa" with a pitch of 200 Hz.
The speech signal contains a strong component at 200 Hz.

Example

Let's consider the following signal:

$$x(t) = 2\cos(100\pi t) + 3\cos(300\frac{\pi}{4}t)$$

- Here, we have two cosine waves with amplitudes 2 and 3 respectively.
- Using the standard cosine equation $\cos(2\pi f t)$, we can find the two frequencies as below:
 - $100\pi = 2\pi f ; f=50\text{Hz}$
 - Similarly, $\frac{300}{4}\pi = 2\pi f ; f= 37.5 \text{ Hz}$
- So, the signal has only two frequencies 50Hz and 37.5Hz.

- Now, convert each cosine into two complex exponential:
- **First cosine:** $2\cos(100\pi t)$ becomes $2\left(\frac{e^{j2\pi 50t} + e^{-j2\pi 50t}}{2}\right)$ using the formula $\cos(\theta) = \left(\frac{e^{j\theta} + e^{-j\theta}}{2}\right)$. Now, as 2 cancels the denominator, we get $e^{j2\pi 50t} + e^{-j2\pi 50t}$
- Similarly, the second cosine $3\cos(2\pi 37.5t)$ becomes

$$\frac{3}{2}e^{j2\pi 37.5t} + \frac{3}{2}e^{-j2\pi 37.5t}$$

- Now, by applying the fourier transform, first cosine produces $\delta(f - 50) + \delta(f + 50)$ and the second cosines produces

$$\frac{3}{2}\delta(f - 37.5) + \frac{3}{2}\delta(f + 37.5)$$

- Putting everything together, the **final fourier transform** is as below:

$$X(f) = \delta(f - 50) + \delta(f + 50) + \frac{3}{2}\delta(f - 37.5) + \frac{3}{2}\delta(f + 37.5)$$

- This signal is made of two cosine waves: one at 50 Hz and another at 37.5 Hz.
- According to Euler's formula, each cosine can be represented by two complex exponentials, one rotating in the positive direction and one in the negative direction.
- Therefore, the Fourier Transform shows four impulses: at +50, -50, +37.5, and -37.5 Hz.
- The original amplitudes of 2 and 3 are split equally between the positive and negative frequency components, giving magnitudes of 1 and 1.5, respectively.
- The Fourier Transform is simply telling us which frequencies are present in the signal and how strong they are."

Properties of Fourier Transform

- **Linearity:** $\mathbf{F}\{\mathbf{ax}_1(\mathbf{t}) + \mathbf{bx}_2(\mathbf{t})\} = \mathbf{aX}_1(\mathbf{f}) + \mathbf{bX}_2(\mathbf{f})$
 - If a signal is made by adding two signals, then its Fourier Transform is simply the sum of their individual Fourier Transforms.
- **Time Shift Property:** $\mathbf{F}\{\mathbf{x}(\mathbf{t} - \mathbf{t}_0)\} = \mathbf{e}^{-j2\pi\mathbf{f}\mathbf{t}_0}\mathbf{X}(\mathbf{f})$
 - If the signal starts later or earlier, the magnitude spectrum remains exactly the same. Only the phase changes.
- **Frequency Shift Property:** $\mathbf{F}\{\mathbf{x}(\mathbf{t})\mathbf{e}^{j2\pi\mathbf{f}_0\mathbf{t}}\} = \mathbf{X}(\mathbf{f} - \mathbf{f}_0)$
 - Multiplying a signal by a sinusoid moves its spectrum along the frequency axis. The shape stays the same. Only the location changes.

- **Scaling Property:** $F\{x(at)\} = \frac{1}{|a|} X\left(\frac{f}{a}\right)$
 - Changing the speed of the signal changes its frequency content.
- **Convolution Property:** $F\{x(t) * h(t)\} = X(f)H(f)$
 - Convolution in the time domain becomes multiplication in the frequency domain.
 - This is one of the most useful properties in signal processing.

Practice Exercises

- For the given signals, find the frequency of each cosine, exponential representation and Fourier Transform:
 - $x(t) = 6\cos(120\pi t) + 4\cos(40\pi t)$
 - $x(t) = 4\cos(2\pi 20t) + 2\cos(2\pi 60t)$
 - $x(t) = 3\cos(2\pi 15t) + 5\cos(2\pi 50t)$
 - $x(t) = 8\cos(2\pi 10t) + 4\cos(2\pi 25t) + 6\cos(2\pi 40t)$

Inverse Fourier Transform

- **Inverse Fourier transform** is used to reconstruct the original signal back from the frequency-domain representation.

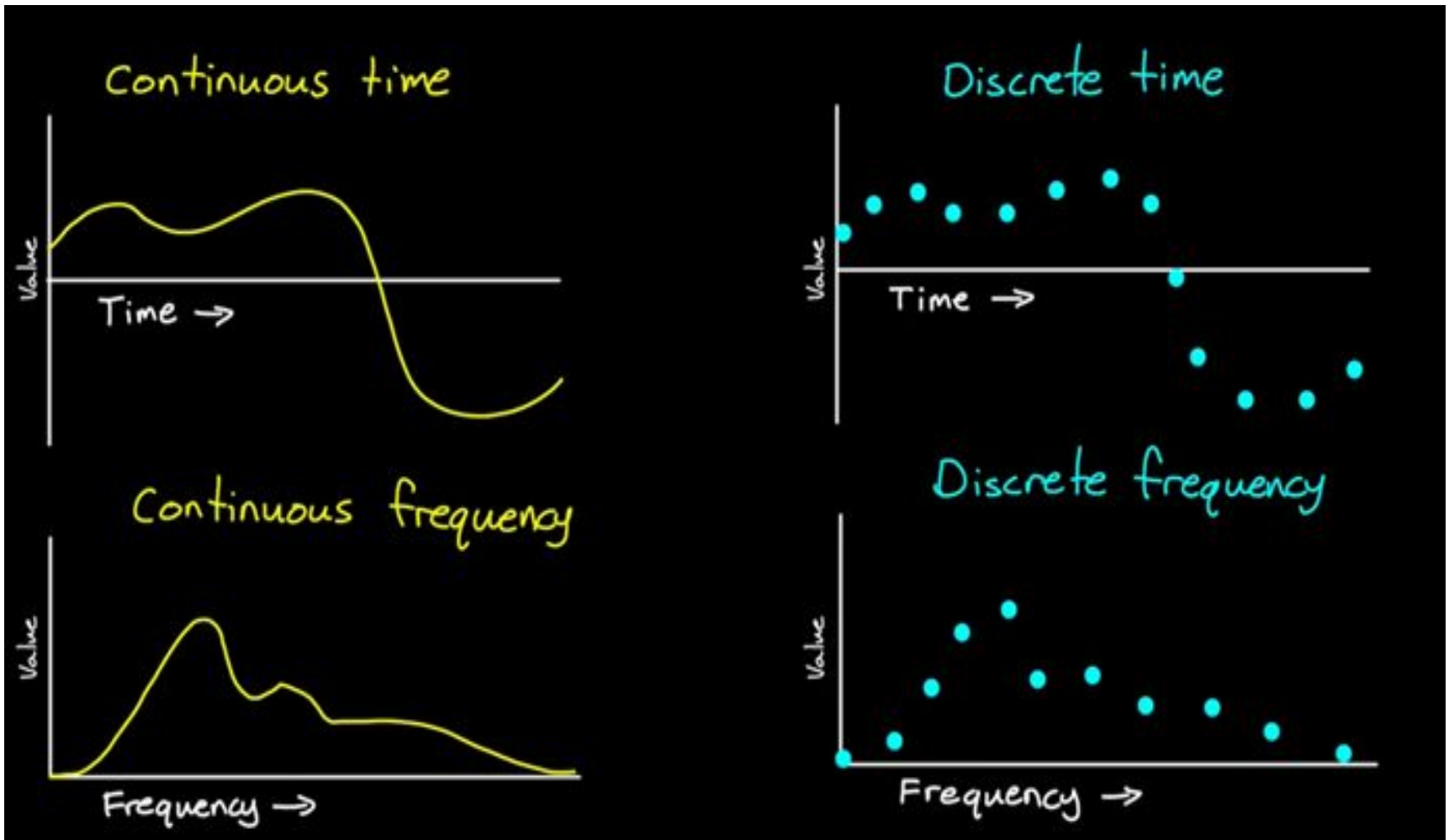
$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df$$

- Using angular frequency $\omega = 2\pi f$,

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

Discrete Fourier Transform

- Continuous and discrete frequency



Discrete-Time Fourier Transform (DTFT)

- For a **discrete-time signal** $x[n]$, Fourier transform decomposes the signal into a sum of sinusoidal with different frequencies.

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

Where, n is **discrete** (samples),

ω (*digital angular frequency*) is **continuous**.

$X(\omega)$ is a **continuous curve**.

- DTFT shows the spectrum at **every possible frequency**.
- In terms of normalized frequency f_{norm} (continuous),

$$X(f) = \sum_{n=-\infty}^{\infty} x[n]e^{-j2\pi f_{norm}n}$$

- **Inverse DTFT**: A large class of sequences can be represented as a linear combination of complex exponentials whose frequencies lie in the range $[-\pi, \pi]$.

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$$

- **Forward DTFT and inverse DTFT** together forms the *discrete-time Fourier transform* pair representation of a sequence.

➤ **Why $-\pi$ to π ?**

- Sinusoidal signals are periodic. The spectrum repeats every 2π .

$$e^{j(\omega+2\pi)n} = e^{j\omega n}$$

- Any interval of length 2π contains all the unique spectral information

Discrete Fourier Transform

- Discrete Fourier Transform (DFT) samples the DTFT at discrete frequencies for finite-length signals.
- DFT is represented as

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

Where, n = time index, k = frequency-bin index, N = number of samples

- DFT is sampled DTFT: $X[k] = X\left(\omega_k = \frac{2\pi k}{N}\right)$. k represents the **DFT frequency bins**.
- For each frequency bin k , the DFT multiplies the signal by a complex exponential. These complex exponentials are called

- **Orthogonality property of DFT basis functions:**

$$\sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}(k-m)n} = \begin{cases} N, & m = k \\ 0, & m \neq k \end{cases}$$

- If a basis function is compared **with itself** ($k = m$), they are perfectly aligned, so the sum is N .
- If **two different basis functions** are compared ($k \neq m$), their positive and negative contributions cancel over one complete period, so the sum is 0.
- Orthogonality guarantees that the DFT cleanly separates frequency components.

Example

Suppose $x[n] = 2\cos\left(\frac{2\pi}{5}n\right) = e^{j2\pi\frac{n}{5}} + e^{-j2\pi\frac{n}{5}}$

Consider $N = 5$, The five samples are $x[n] = [2\cos 0, 2\cos \frac{2\pi}{5}, 2\cos \frac{4\pi}{5}, 2\cos \frac{6\pi}{5}, 2\cos \frac{8\pi}{5}] = [2, 0.618, -1.618, -1.618, 0.618]$

Digital angular frequencies: $0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5} = -\frac{4\pi}{5}, \frac{8\pi}{5} = -\frac{2\pi}{5}$

5 bins for 5 samples, $X[k] = \sum_{n=0}^4 x[n]e^{-j\frac{2\pi}{5}kn}$

5 frequency bins are $x[0], x[1], x[2], x[3], x[4]$

- $X[0] = 2 + 0.618 - 1.618 - 1.618 + 0.618 = 0$

- $$X[1] = \sum_{n=0}^4 \left(e^{j2\pi\frac{n}{5}} + e^{-j2\pi\frac{n}{5}} \right) e^{-j2\pi\frac{n}{5}}$$

$$= \sum_{n=0}^4 1 + \sum_{n=0}^4 e^{-j4\pi\frac{n}{5}} = 5$$

- $$X[2] = \sum_{n=0}^4 \left(e^{j2\pi\frac{n}{5}} + e^{-j2\pi\frac{n}{5}} \right) e^{-j4\pi\frac{n}{5}}$$

$$= \sum_{n=0}^4 e^{-j2\pi\frac{n}{5}} + \sum_{n=0}^4 e^{-j6\pi\frac{n}{5}} = 0$$

- $X[3] = 0$

- $X[4] = 5$ (same as bin 1)

[Note: $\sum_{n=0}^4 e^{-j4\pi\frac{n}{5}}$, $\sum_{n=0}^4 e^{-j6\pi\frac{n}{5}}$, $\sum_{n=0}^4 e^{-j2\pi\frac{n}{5}}$ value will be zero due to **orthogonality property of exponentials**]

- The DFT acts like a bank of frequency detectors. When the frequency

Example

Suppose a sine wave $x(t) = \sin(2\pi ft)$ with 1Hz frequency, amplitude 1, sampling frequency: 8 Hz. Apply DFT.

Solution:

Sampling frequency $f_0 = 8 \text{ Hz}$,

Sampling period $t_s = 1/8 \text{ sec}$

i.e., we take sample every 1/8 seconds.

Hence, Sampling instant $t = n/8$

$$\text{So, } x[n] = A \sin(2\pi f_0 t) = 1 \cdot \sin\left(2\pi \cdot 1 \cdot \frac{n}{8}\right) = \sin\left(\frac{\pi}{4} n\right)$$

Frequencies: $0, \pi/4, (\pi/2), (3\pi/4), (\pi), (5\pi/4), (6\pi/4), (7\pi/4)$

Sampled signal $x[n]$

- N=8 means we have **8 frequency bins** separated by **1Hz** and at bin 1 we have the frequency 1Hz. i.e., the first peak (**1Hz**) should be at **bin 1** and as it is sine wave, we will have another corresponding **peak (-1Hz) at bin 7**.

- N-point DFT is $X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$

- For N=8, $X[k] = \sum_{n=0}^7 x[n] e^{-j\frac{2\pi}{8}kn}$

- Frequency spacing of DFT is $\Delta f = \frac{F_s}{N} = 1/1 = 1Hz$

- For $k=1$, $X[1] = \sum_{n=0}^7 x[n] e^{-j2\pi\frac{n}{8}}$

$$= 0 \cdot e^{-j0} + 0.707e^{-j\frac{\pi}{4}} + 1e^{-j\frac{\pi}{2}} + 0.707e^{-j\frac{3\pi}{4}} + 0 \cdot e^{-j\pi} - 0.707e^{-j\frac{5\pi}{4}} - 1 \cdot e^{-j\frac{6\pi}{4}} - 0.707e^{-j\frac{7\pi}{4}}$$

- DFT is **multiplying the signal** by a frequency component corresponding to **1 Hz**.

- Now simplify the complex terms using **euler's formula**

$$e^{-j\theta} = \cos\theta - j\sin\theta$$

$$- 0 \cdot e^{-j0} = 0 \cdot \cos 0 - j \sin 0 = 0$$

$$- 0.707e^{-j\frac{\pi}{4}} = 0.707(\cos\frac{\pi}{4} - j\sin\frac{\pi}{4}) = \\ 0.707(0.707 - j0.707) = \mathbf{0.5 - j0.5}$$

$$- 1e^{-j\frac{\pi}{2}} = 1(\cos\frac{\pi}{2} - j\sin\frac{\pi}{2}) = 1(0 - j1) = -j$$

- $0.707e^{-\frac{j3\pi}{4}} = 0.707(\cos\frac{3\pi}{4} - j\sin\frac{3\pi}{4}) = 0.707(-0.707 - j0.707) = -0.5 - j0.5$
- $0.e^{-j\pi} = 0$
- $-0.707e^{-\frac{j5\pi}{4}} = -0.707(\cos\frac{5\pi}{4} - j\sin\frac{5\pi}{4}) = -0.707(-0.707 + j0.707) = 0.5 - j0.5$
- $-1.e^{-j\frac{6\pi}{4}} = -(\cos\frac{3\pi}{2} - j\sin\frac{3\pi}{2}) = -(0 - j(-1)) = -j$
- $-0.707e^{-j\frac{7\pi}{4}} = -0.707(\cos\frac{7\pi}{4} - j\sin\frac{7\pi}{4}) = -0.707(0.707 + j0.707) = -0.5 - j0.5$

- Now, putting all the terms together, we get

$$x[1] = 0 + (0.5 - j0.5) + (-j) + (-0.5 - j0.5) + 0 + (0.5 - j0.5) + (-j) + (-0.5 - j0.5) = -4j$$

- *Now, the magnitude and phase*

- **Magnitude**, $|x[1]| = \sqrt{0^2 + (-4)^2} = 4$

- **Phase is -90°**

- The sine wave's energy is concentrated at **bin 1 (and symmetrically at bin 7)**, confirming the 1 Hz frequency component.

- $X[1] = -4j \rightarrow$ magnitude 4, phase -90°

- Why the **magnitude is 4** when **amplitude is 1**???
- For a sinusoid whose frequency exactly matches a DFT bin:
- $|x[k]| = \frac{AXN}{2}$; where, A is amplitude and N is no. of samples
- Here, $|x[1]| = \frac{1 \times 8}{2} = 4$
- The DFT acts like a set of **frequency detectors**.
- Since our signal is **1 Hz** and **bin 1 represents 1 Hz**, **bin 1** produces a **strong output**. The other frequency components cancel out.

– Note: The DFT procedure is exactly the same for sine and cosine signals.

Sine wave: 1Hz, Amplitude = 1

Sampling Frequency: 8 Hz

samples (N): 8

Fourier Coefficients:

$$X_0 = 0$$

$$X_1 = 0 - 4j$$

$$X_2 = 0$$

$$X_3 = 0$$

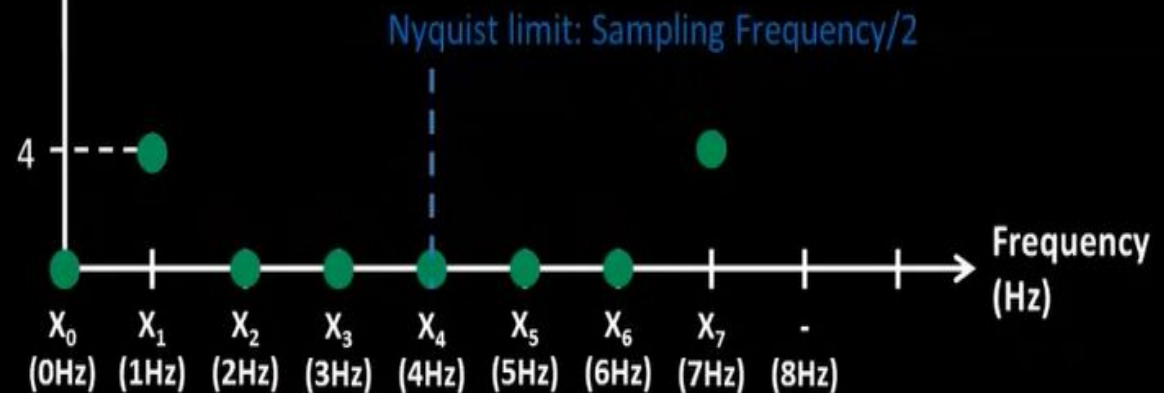
$$X_4 = 0$$

$$X_5 = 0$$

$$X_6 = 0$$

$$X_7 = 0 + 4j$$

$$Mag = \sqrt{A_k^2 + B_k^2} = \sqrt{0^2 + (-4)^2} = 4$$



- In the previous example, the sine wave's energy is concentrated at **particular bins**.
- If the bins align correctly with the sampling frequency, DFT shows clear spikes.
- **Clear spikes** at the matching bins (positive and negative frequency components). **Zero elsewhere**, because the orthogonality of exponentials cancels out all other bins.
- If the signal frequency does **not** align exactly with a bin, energy spreads across multiple bins (nonzero complex values in several bins). This is called **spectral leakage**.

- DFT is

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

where,

N number of **basis functions**.

N^2 multiplications and additions.

So complexity of DFT is

$$O(N^2)$$

If $N = 1024$, ~1 million operations are required.

- A sine wave has: Frequency: $f_0=1$ Hz, Amplitude: $A=1$
Sampling frequency: $F_s=16$ Hz. Apply an 8-point DFT to the sampled signal.
- A sine wave has: Frequency: $f_0=2$ Hz, Amplitude: $A=1$
Sampling frequency: $F_s=16$ Hz. Apply an 8-point DFT to the sampled signal.
- A sine wave has: Frequency: $f_0=3$ Hz, Amplitude: $A=2$
Sampling frequency: $F_s=12$ Hz. Apply an 5-point DFT to the sampled signal.

Fast Fourier Transform

- Popularized by **James W. Cooley** and **John W. Tukey** in 1965.
- Uses **divide and conquer** approach to reduce **computational complexity of DFT**.
- Uses **Even-Odd decomposition** to form smaller subproblems.
- By exploiting the **symmetry and periodicity** of the complex exponential terms, known as **twiddle factors**, the FFT avoids **redundant** computations and **reuses intermediate results**.
- FFT **reduces** computational complexity of DFT $O(N^2)$ to $O(N \log_2 N)$
- *FFT is not a new transform; it is simply a faster algorithm for calculating the DFT.*

Even-Odd Decomposition

- The **N-point DFT** is $X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn}$
- Instead of considering all n together, divide the samples into:
 - Even indices; $n=0,2,4,6,\dots$
 - Odd indices; $n=1,3,5,7,\dots$
- For example, for **N=8**: $x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7$
 - Even indices: x_0, x_2, x_4, x_6
 - Odd indices: x_1, x_3, x_5, x_7

$$F_k = \sum_{m=0}^{N/2-1} x_{2m} \cdot e^{-\frac{j2\pi k(2m)}{N}} + \sum_{m=0}^{N/2-1} x_{2m+1} \cdot e^{-\frac{j2\pi k(2m+1)}{N}}$$

- Let $n = 0, 1, 2, 3, \dots$ then:
 - **$2m = 0, 2, 4, 6, \dots$** ; Exactly the **even indices**.
- Similarly, **$2m + 1 = 1, 3, 5, 7, \dots$** ; Exactly the **odd indices**.
- So the complicated-looking equation is simply saying:
 - **DFT = DFT of even samples + DFT of odd samples**
- **Even part:** N-point DFT $\rightarrow$ N/2-point DFT of even samples

$$\sum_{m=0}^{N/2-1} x_{2m} e^{-j \frac{2\pi km}{N/2}}$$

- **Odd part:**

$$\sum_{m=0}^{N/2-1} x_{2m+1} e^{-j \frac{2\pi k(2m+1)}{N}}$$

- Simplify the exponential as below:

$$\begin{aligned}
 e^{-j\frac{2\pi k(2m+1)}{N}} &= e^{-j\frac{2\pi k(2m)}{N}} e^{-j\frac{2\pi k}{N}} \\
 &= e^{-j\frac{4\pi km}{N}} e^{-j\frac{2\pi k}{N}} \\
 &= e^{-j\frac{4\pi km}{2(N/2)}} e^{-j\frac{2\pi k}{N}} \\
 &= e^{-j\frac{2\pi km}{N/2}} e^{-j\frac{2\pi k}{N}}
 \end{aligned}$$

- Now, substitute this value in the odd part exponential, we get

$$\sum_{m=0}^{N/2-1} x_{2m+1} e^{-j\frac{2\pi km}{N/2}} e^{-j\frac{2\pi k}{N}}$$

- The second factor $e^{-j\frac{2\pi k}{N}}$ does not depend on m. Hence, the equation can be written as below:

$$e^{-j\frac{2\pi k}{N}} \sum_{m=0}^{N/2-1} x_{2m+1} e^{-j\frac{2\pi km}{N/2}}$$

- This factor is called **Twiddle Factor** and represented as below

$$W_N^k = e^{-j\frac{2\pi k}{N}}$$

- Now, if the even sample is defined as $\mathbf{E}[\mathbf{k}]$ and odd sample is defined as $\mathbf{O}[\mathbf{k}]$, then the original DFT becomes

$$\mathbf{X}[\mathbf{k}] = \mathbf{E}[\mathbf{k}] + W_N^k \mathbf{O}[\mathbf{k}]$$

- So, instead of taking **one 8-point DFT**, we can take one **4-point DFT of even samples** and another **4-point DFT of odd samples** and then **combine** them using the **twiddle factor**.
- **Twiddle factor:** It is the complex exponential used in the FFT to **rotate/weight** the frequency components while combining the **even and odd parts**.
 - It helps **combine** the smaller DFTs efficiently to obtain the final DFT, making FFT **much faster** than direct DFT calculation.
- **Note:** The FFT does not change the DFT result.
 - It simply calculates the DFT more efficiently. First, we separate the input samples into even-indexed and odd-indexed samples. This gives two smaller DFTs.
 - We then multiply the odd part by a twiddle factor and combine the even and odd parts. The same process is repeated on the smaller DFTs.

Butterfly Network in FFT

- A **butterfly** is the basic calculation unit of an FFT.
- It takes two input values, combines them using a twiddle factor, and produces two output values.

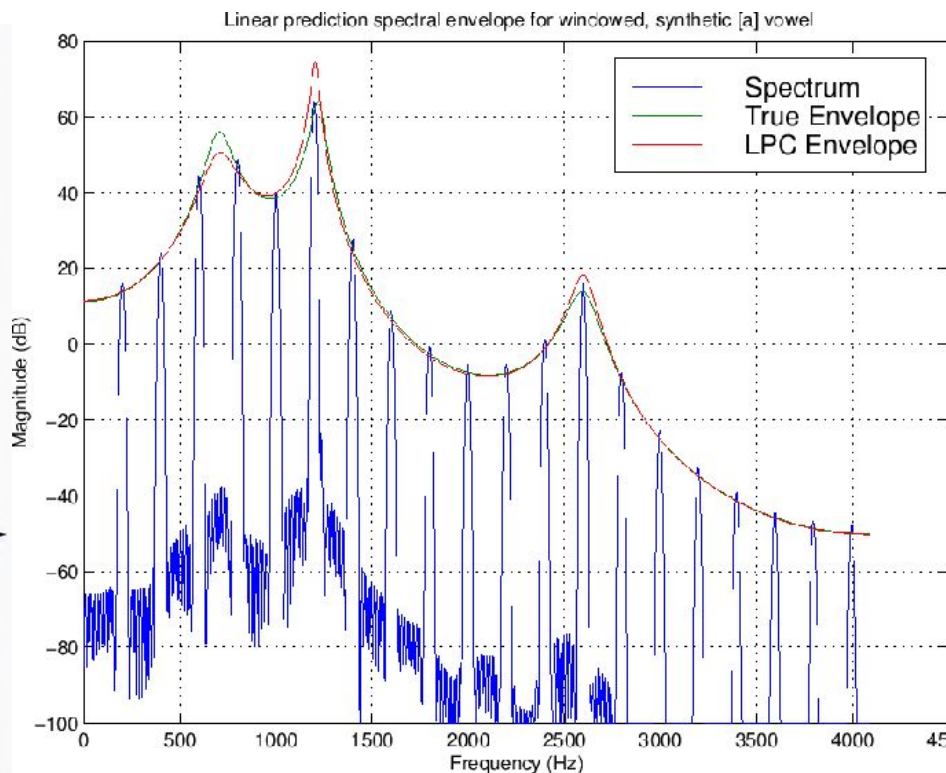
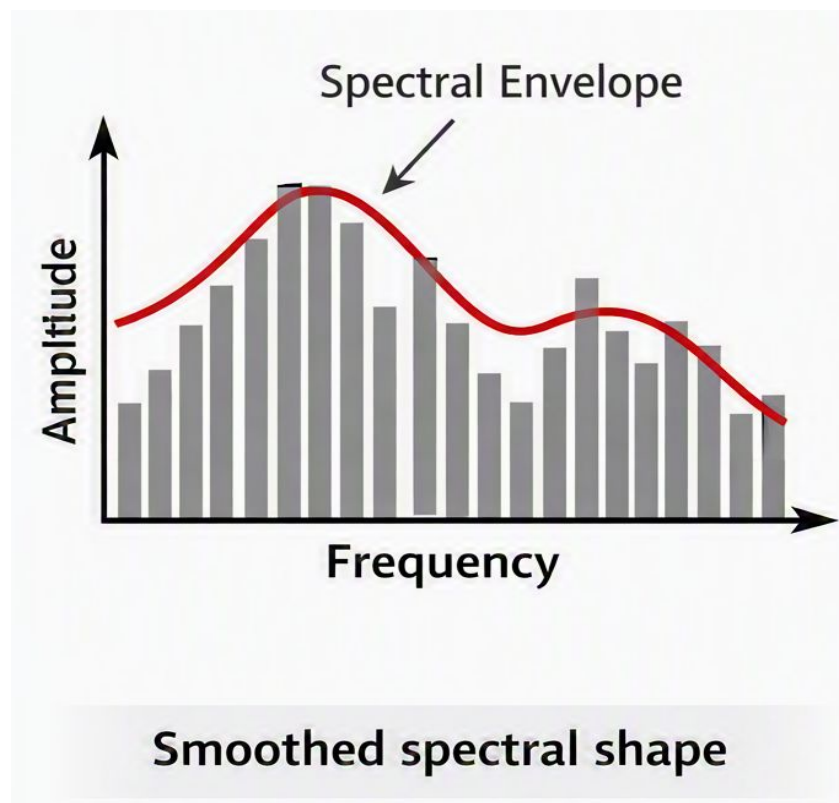
$$X1 = A + WB \text{ and } X2 = A - WB;$$

where, W is the twiddle factor

- A butterfly takes two values, performs one addition and one subtraction after applying a twiddle factor, and produces two outputs.
- FFT uses many butterfly operations to break a large DFT into smaller calculations, making the computation much faster.

Spectral Analysis Models

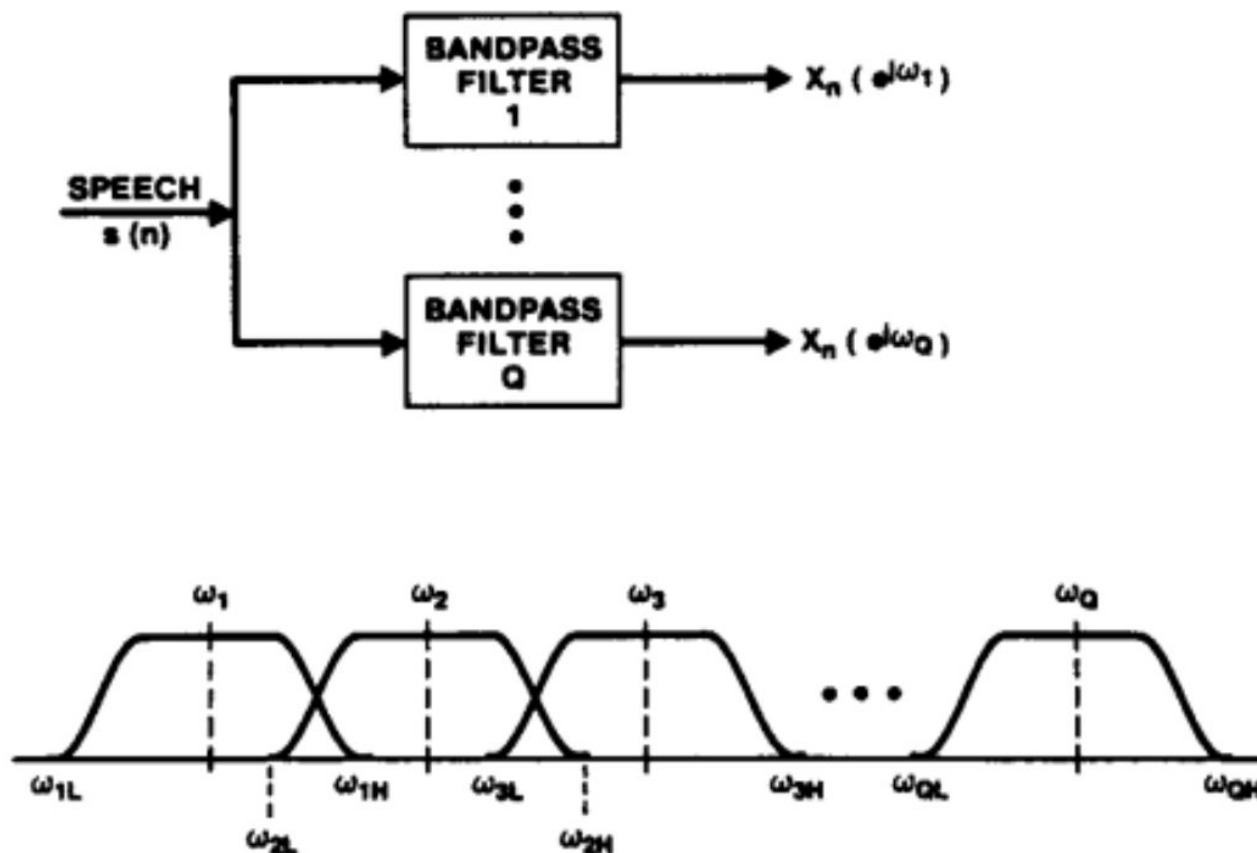
- The **spectral envelope** is the smooth curve outlining the peaks and valleys of the speech spectrum, carries crucial information about **phonemes** and **speaker identity**.



- In **speech recognition systems**, the signal-processing front end is a crucial component that converts the speech waveform into a parametric representation for further analysis.
- One of the most important parametric representation is the *short-time spectral envelope*.
- Two dominant methods for this are:
 - the **bank-of-filters** model
 - the **linear predictive coding** (LPC) model.
- Both methods are used in the initial “parameter measurement” phase of either the pattern recognition or the acoustic-phonetic approach to speech recognition.

Bank of Filter (BOF) Model

- The **bank-of-filters** approach performs spectral analysis by passing the speech signal through a set of **bandpass filters** that span the frequency range of interest.



- **Speech signal** is passed through multiple **bandpass filters** in parallel.
- Each **filter** covers a specific **frequency band** and allows only that range of frequencies **to pass**.
- The filters are usually **overlapping**, so the complete speech frequency range is covered without missing important information.
- The **output of each filter** represents the **energy** present in that particular **frequency band**.
- Together, all filter **outputs** provide a **spectral** representation of the speech signal, which is useful for speech analysis and recognition.

Types of Digital Filters

☐ Uniform Filter Banks:

- ☐ The filters are spaced **uniformly** across the frequency range, having **equal bandwidth**.
- ☐ The number of filters, Q , should not be too small (less than 8) to resolve the spectrum adequately, nor too large (more than 32) as it might result in some bands having very low energy for certain speakers.
- ☐ Doesn't match human auditory perception.

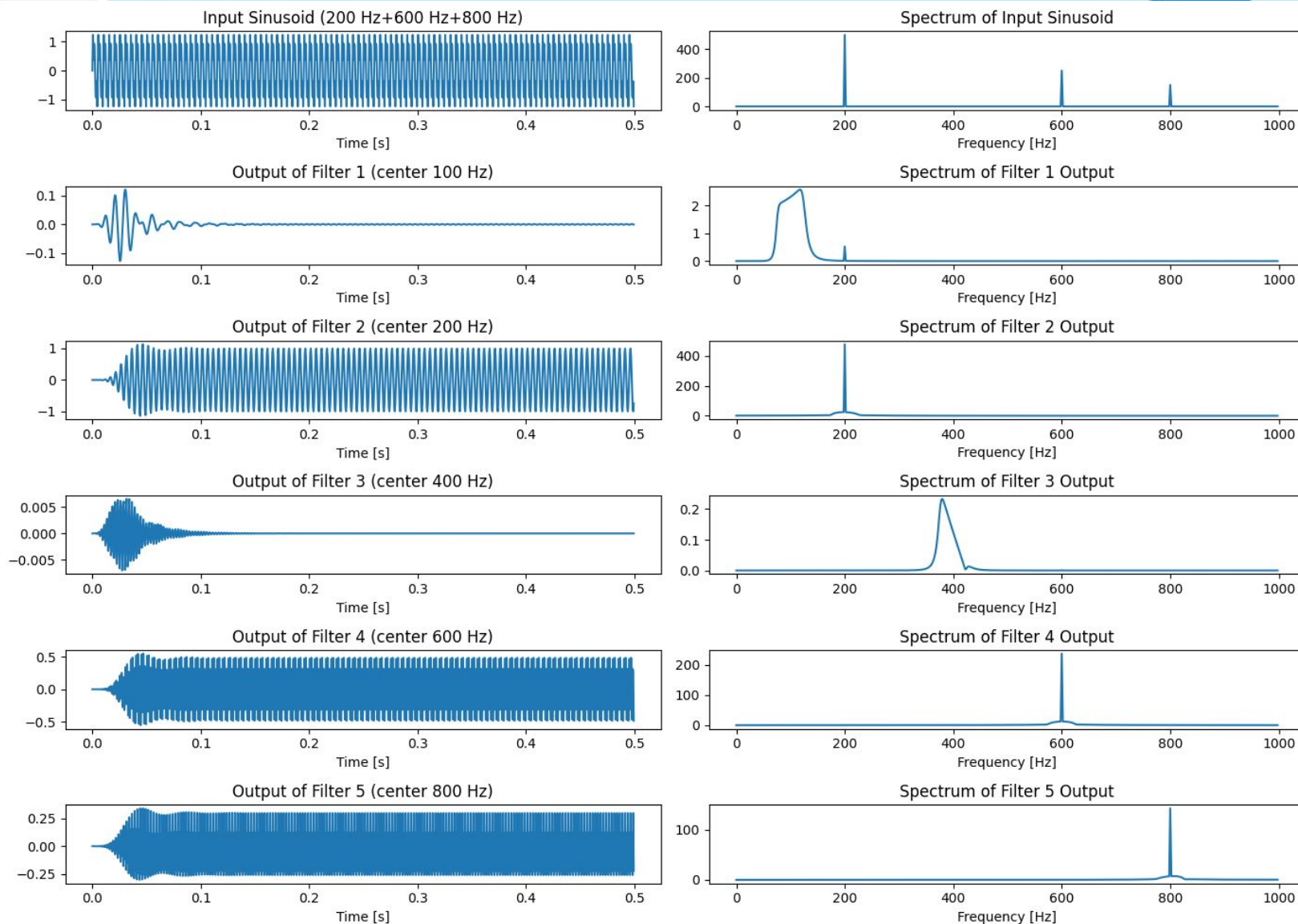
☐ Non-Uniform Filter Banks:

- ☐ Filters with **varying bandwidths**, often narrower at low frequencies and wider at high frequencies
- ☐ These are designed based on criteria related to human auditory perception.
- ☐ Complex and computationally heavier than uniform banks₄₅

- **FIR vs. IIR Filters:** Filter banks can be implemented using either **Infinite Impulse Response (IIR)** or **Finite Impulse Response (FIR)** filters.
- **FIR filters** are stable.
 - The output is dependent on only current and past inputs,
 - They can achieve linear phase (waveform shape is preserved), and
 - Their computational cost is significantly reduced using Fast Fourier Transform (FFT) implementations.
- A highly efficient way to implement a uniform FIR filter bank is through the **STFT**. By modulating a fixed lowpass window, the outputs of the filter bank can be computed using an FFT algorithm (polyphase filter-bank structure).

- ❑ **Tree Structures:** Non-uniform filter banks, like octave-spaced designs, can be efficiently implemented using a tree structure of Quadrature Mirror Filters (QMFs) with successive stages of filtering and sample rate reduction.
 - ❑ A tree-structured filter bank repeatedly splits the frequency range into smaller bands using QMFs, while downsampling at each stage to make the system computationally efficient.
- ❑ **Low-pass filter:** passes low frequencies, attenuates high.
- ❑ **High-pass filter:** passes high frequencies, attenuates low.
- ❑ **Band-pass filter:** passes a specific frequency band.
- ❑ **Notch filter:** rejects a narrow band, passes others.
- ❑ **All-pass filter:** passes all frequencies but alters phase.

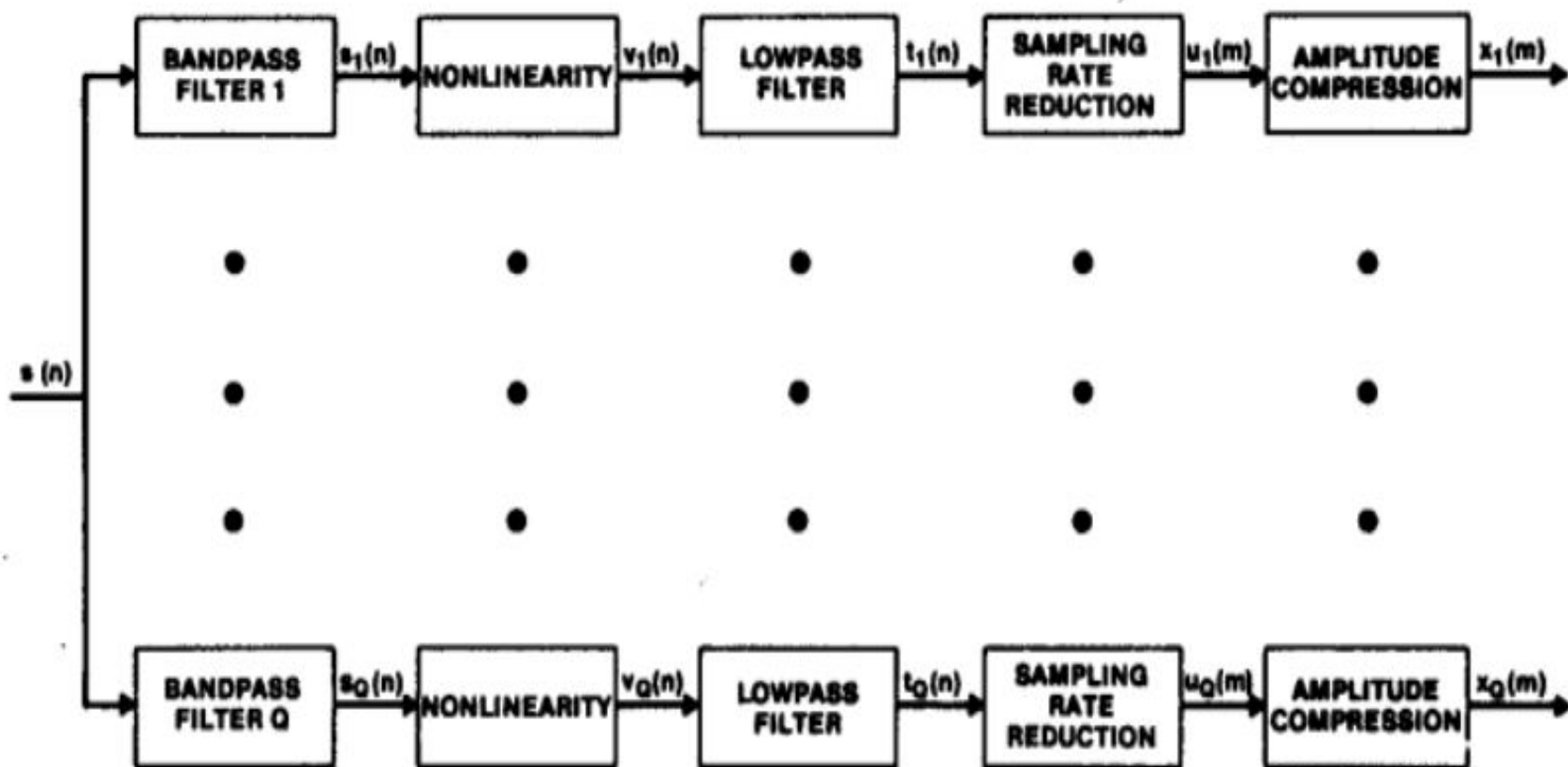
Sample Illustration



- The input is: $x(t) = \sin(2\pi 200t) + \sin(2\pi 600t) + \sin(2\pi 800t)$; So the input contains three frequency components: **200 Hz, 600 Hz, and 800 Hz.**
- Now, individual frequency components are separated
 - Filter 1 $\rightarrow$ No significant component
 - Filter 2 $\rightarrow$ 200 Hz
 - Filter 3 $\rightarrow$ No significant component
 - Filter 4 $\rightarrow$ 600 Hz
 - Filter 5 $\rightarrow$ 800 Hz
- A filter bank separates a complex signal into its individual frequency components by allowing different frequency bands to pass through different filters.
- The time-domain plots show what each filter produces, while the spectrum plots show which frequency component that filter has

Complete BOF Model Structure

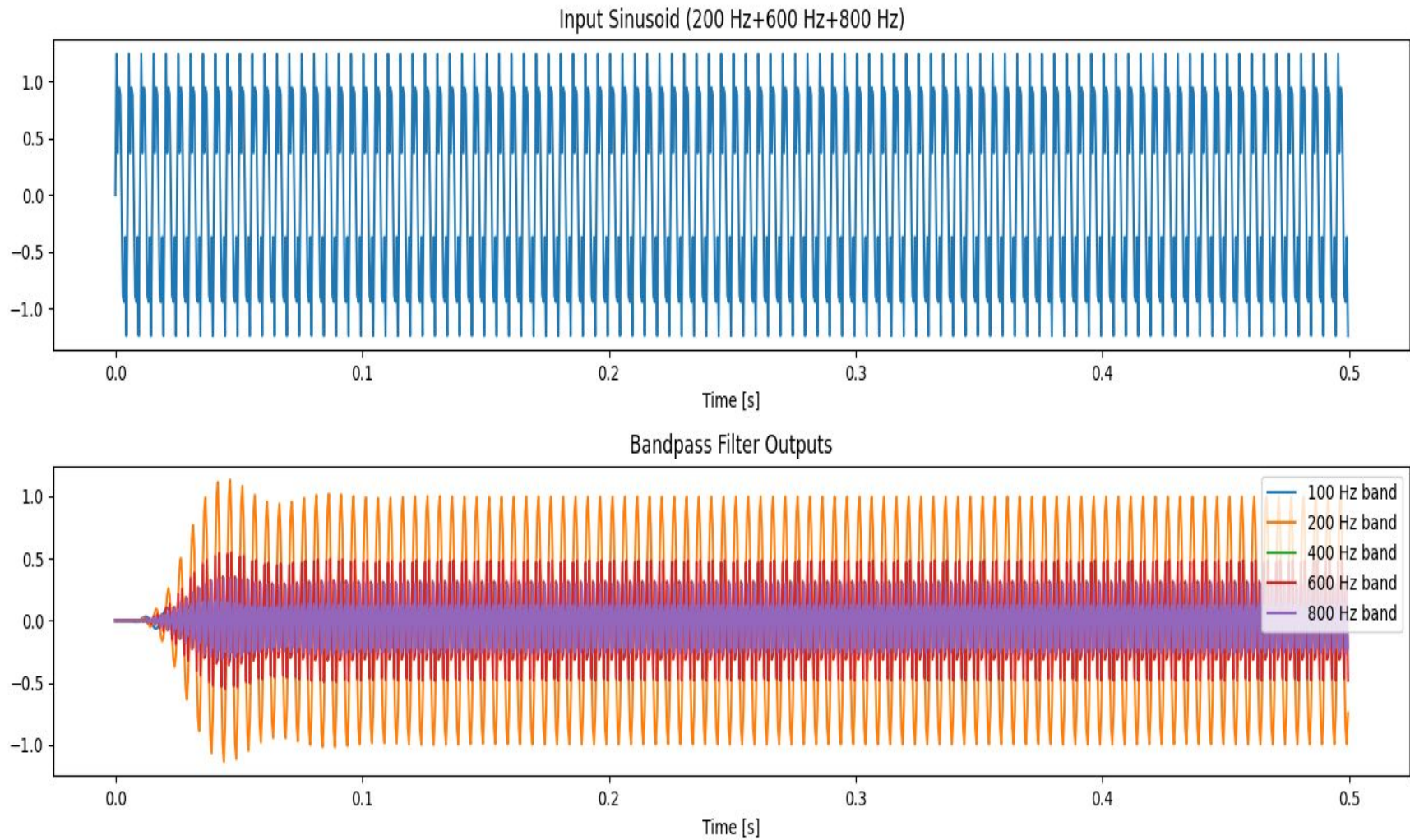
- A complete bank-of-filters front-end analyzer follows a canonical structure, as shown in Figure.



• **Bandpass Filtering:**

- The sampled **speech signal**, $s(n)$, is passed through a bank of **Q bandpass filters**.
- Each filter, $h_i(n)$, has its own **impulse response** and covers a specific **frequency band**.
- It allows signals within a defined range (between a lower and upper cutoff frequency) **to pass**.
- The filters can and generally do **overlap** in frequency.
- The output of the **i^{th} filter**, a bandpass-filtered signal, is denoted as $s_i(n)$.

contd..



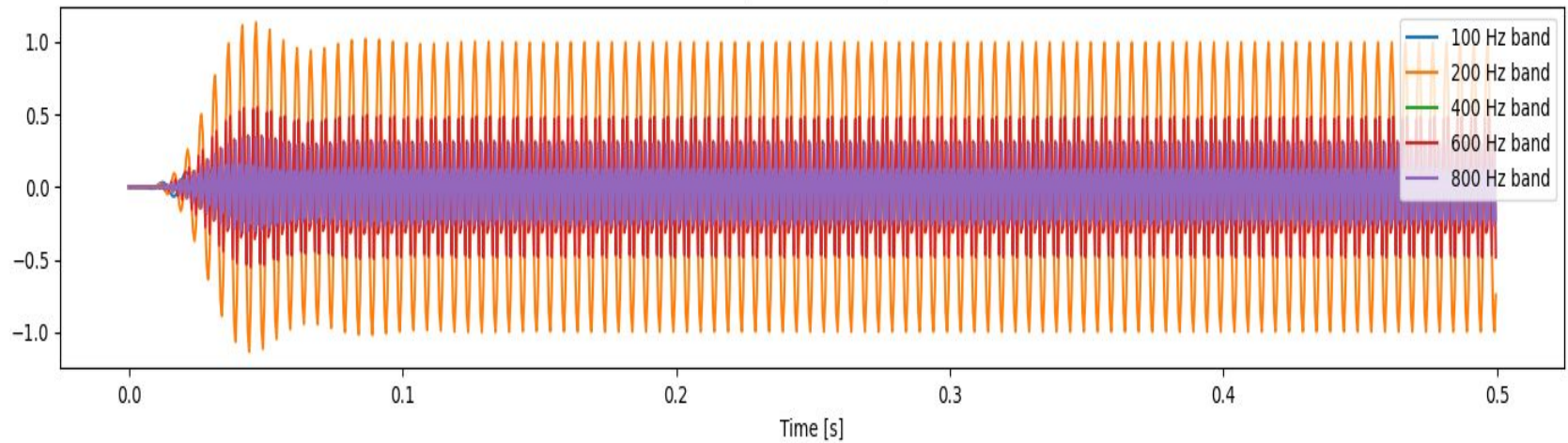
• Nonlinearity (Rectification):

- To measure the energy in each band, the bandpass signals $S_i(n)$ are next passed through a nonlinearity, typically a full-wave or half-wave rectifier.
- This process shifts the signal's **spectrum** to the **low-frequency band** while creating **high-frequency images**.
 - *Full-wave rectifier* takes the absolute value of the signal, flipping negative parts positive.
 - *Half-wave rectifier* sets negative parts to zero, keeping only the positive half.
- Rectification produces:
 - High-frequency harmonics/images ($2f$, $4f$, $6f$...) that are artifacts.

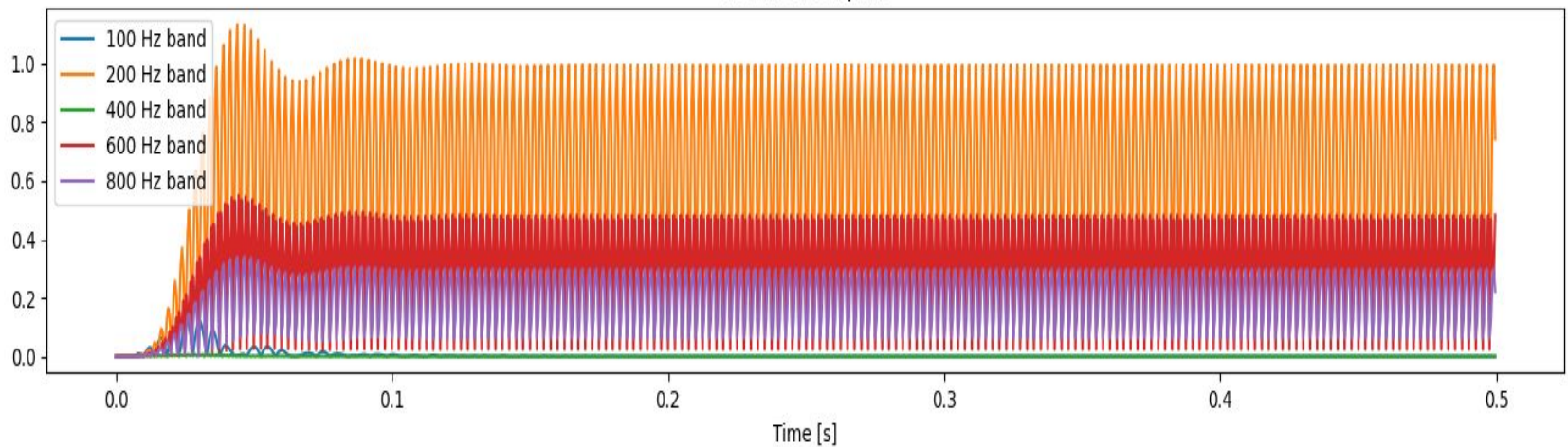
Figure 10.10: Rectification of a bandpass signal

contd..

Bandpass Filter Outputs

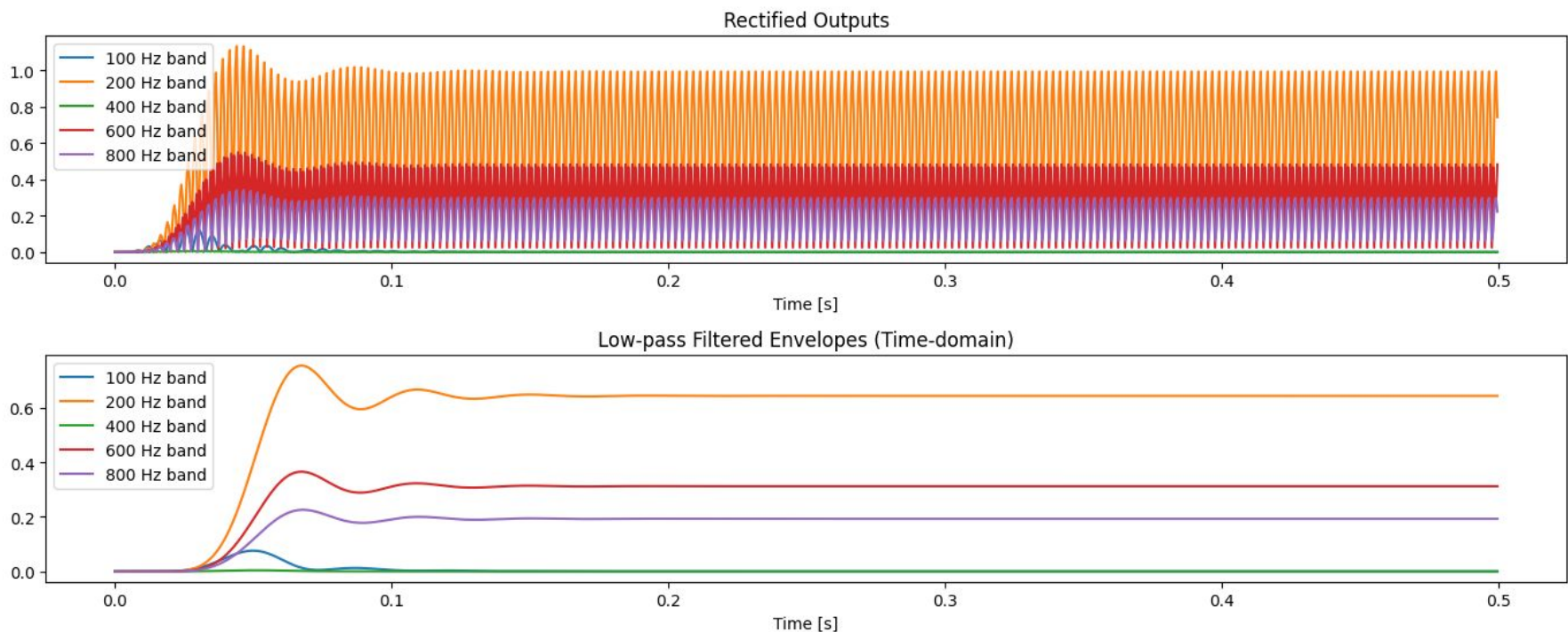


Rectified Outputs



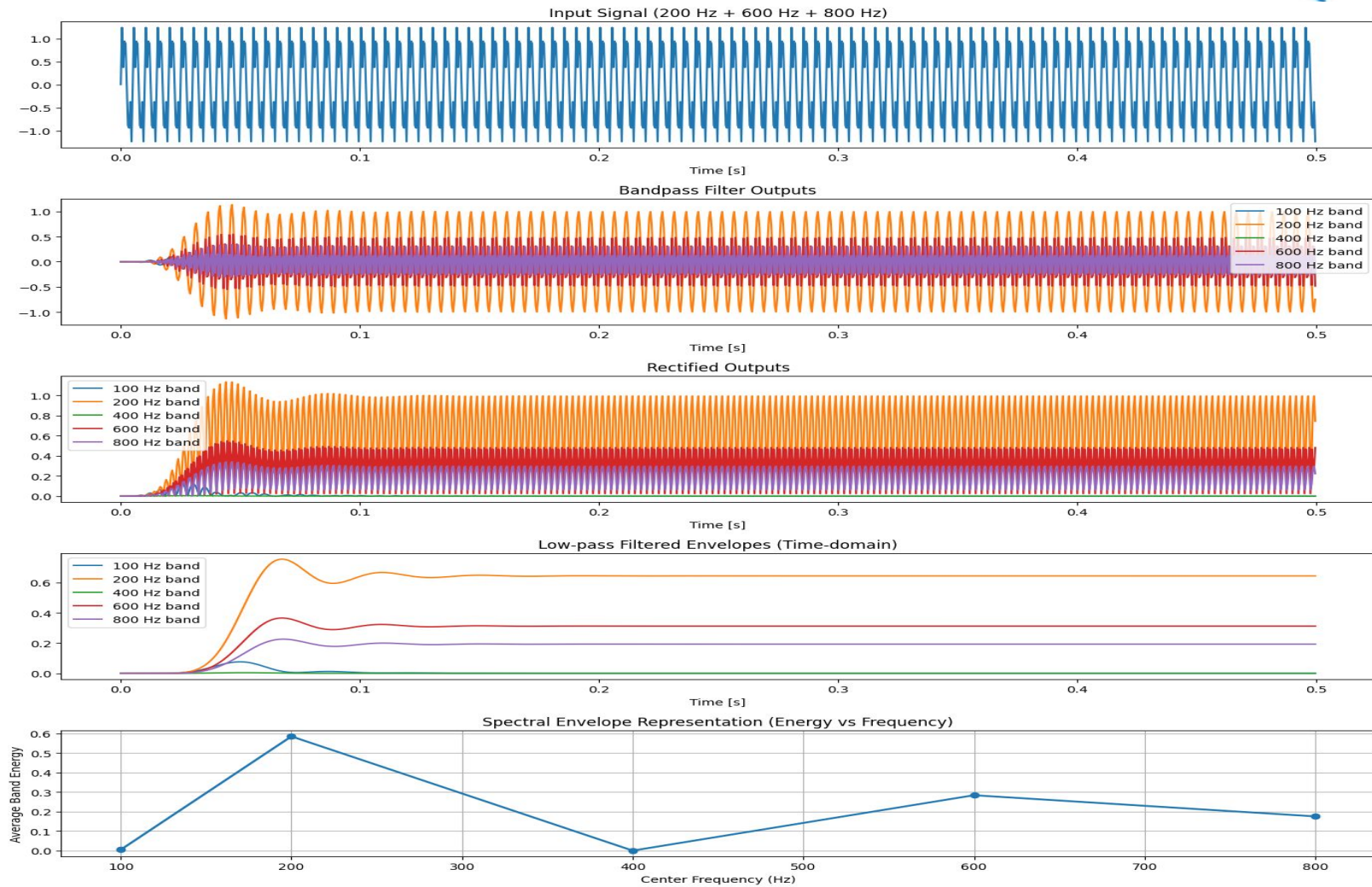
- **Lowpass Filtering:**

- A **lowpass filter** is applied to the output of the nonlinearity to **smooth the signal** and **eliminate** the newly created **high-frequency images**, retaining the DC component that represents an estimate of the **speech signal energy** in that frequency band.



- **Sampling Rate Reduction (Decimation):** The **bandwidth** of the resulting **lowpass signal $u_i(n)$** is related to the rate of change of speech harmonics and is typically around **20–30 Hz**. Therefore, for efficiency, the signal is **resampled** at a much **lower rate**, on the order of **40–60 Hz**.
- **Amplitude Compression:** The dynamic range of the signal is compressed using a **logarithmic or μ -law encoding scheme**.
- This process greatly reduces the speech data rate.
 - **For example**, a speech signal with a **raw bit rate of 240 kbps** can be reduced to about **5.6 kbps** using a 16-channel filter bank and compression. This gives nearly **40:1 data reduction**.
 - However, the important characteristics of the speech signal gets retained.
- Instead of sending the **complete speech waveform** we extract its

contd..



Sample Example

- A narrowband speech signal is sampled at 8 kHz with 8-bit precision. Suppose this signal is processed through a 10-channel filter bank, where each channel output is sampled at 100 Hz and quantized with 6-bit precision.
 - Compute the raw bit rate of the original signal.
 - Compute the compressed bit rate after filter bank processing.
 - Determine the compression ratio achieved.

Soln: Given,

Sampling frequency, $F_s = 8\text{Khz i. e., } 8000\text{sample/sec}$

Precision: 8 bits/sample; Precision after processing: 6 bits/sample

i. **Raw bit rate, $R_{org} = F_s \times \text{precision}$**
 $= 8000 \times 8 = 64000$ i. e., 64kbps

ii. Now, each channel produces **$100 \times 6 = 600$ bits/sec**

So, for 10 channels,

$$\mathbf{R_{compressed} = 10 \times 600 = 6000 \text{ i. e., 6kbps}}$$

iii. **Compression Ratio** $= \frac{R_{org}}{R_{compressed}} = 64 / 6$
 $= 10.67:1$

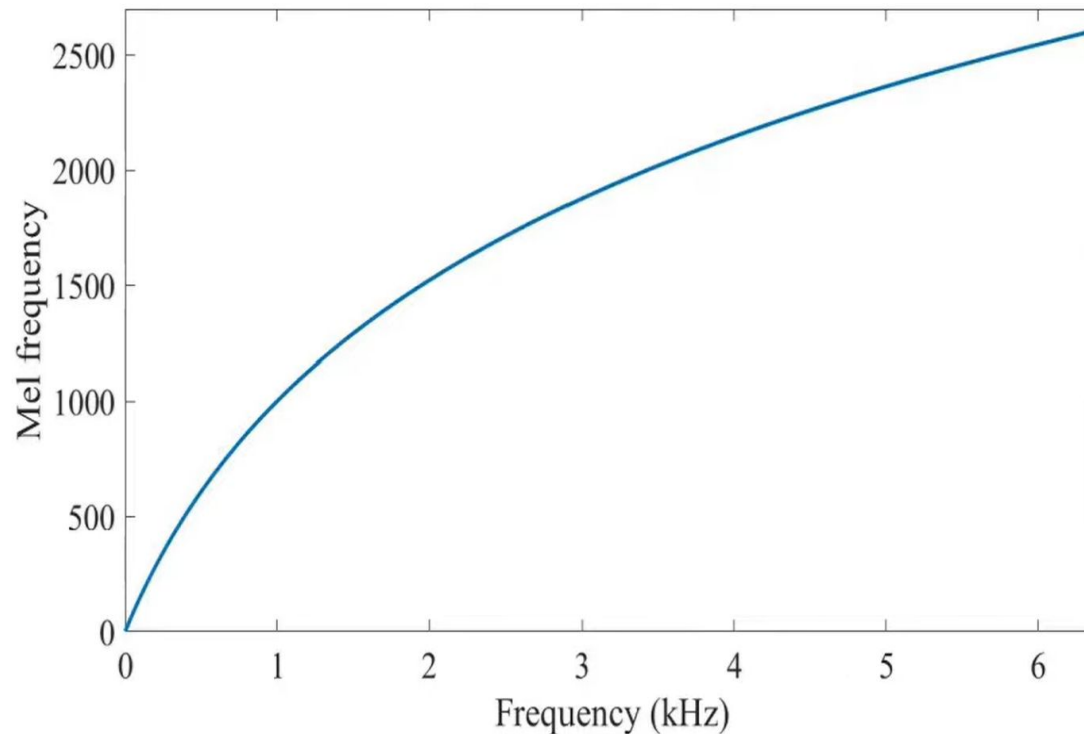
- A speech signal is sampled at 16 kHz with 10-bit precision, giving a raw bit rate of 160 kbits/sec. The signal is processed through an 8-channel filter bank, where each channel output is sampled at 200 Hz and quantized with 6-bit precision. Calculate:
 - The raw bit rate of the original speech signal.
 - The compressed bit rate after filter bank processing.
 - The compression ratio achieved.
- A speech signal is sampled at 12 kHz with 8-bit precision. It is processed through a 12-channel filter bank, where each channel output is sampled at 100 Hz and represented using 5-bit precision. Calculate:
 - The raw bit rate of the original speech signal.
 - The compressed bit rate after filter bank processing.
 - The compression ratio achieved.

Non-Uniform Filters - Mel Filter

- *“Humans perceive frequency logarithmically”*
- Filters can be spaced according to the **critical band scale** (or related scales like the **mel scale** or **bark scale**), based on perceptual studies of how different frequency bands contribute to speech articulation.
- **Mel-scale:** Mel scale converts ordinary frequency in Hz into a scale that approximately represents human perception of pitch.

$$m = M(f) = 2595 \cdot \log_{10} \left(1 + \frac{f}{700} \right),$$
$$f = 700 \left(10^{\frac{m}{2595}} - 1 \right)$$

Where, f = frequency in Hz; m = corresponding mel value



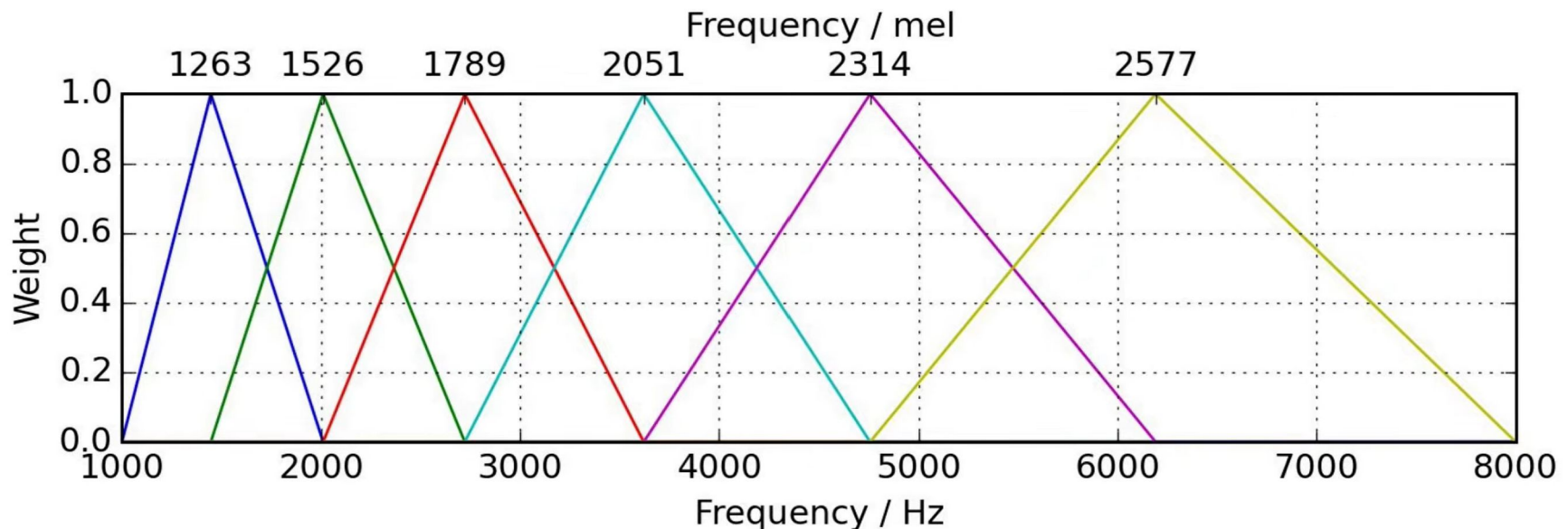
- The curve is almost straight at low frequencies while it becomes increasingly flattened at high frequencies
- This means:
 - Low frequencies are represented with greater resolution, while high frequencies are represented with lower resolution

How to implement???

1. Convert the **lowest** and the **highest** frequency to **Mel**
2. Create **equally spaced points** on the **Mel scale**. These equally spaced Mel points will correspond to **unequally spaced frequencies** in Hz.

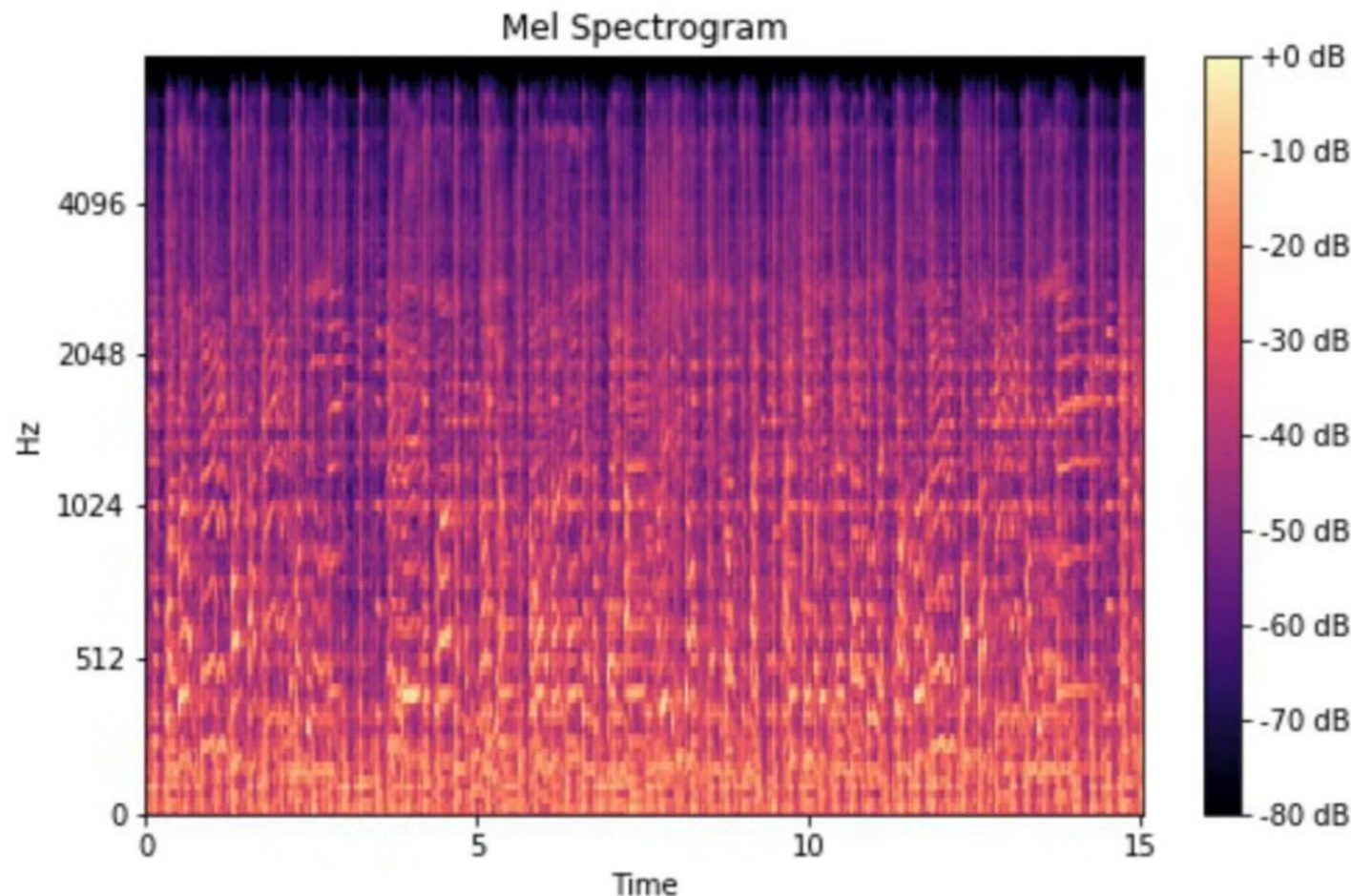
$(\text{Mel}_{\text{high}} - \text{Mel}_{\text{low}}) / (N + 2 - 1)$; where N is triangular filters

3. Convert points back to Herz; $f = 700(10^{\frac{m}{2595}} - 1)$
4. Approximate the frequency bins and create triangular filters



Mel Spectrogram

- After application of Mel Filter Bank, Mel Spectrogram is generated.



Applications of Filter Bank Model

- **Speech Recognition:** Mel filters are used to extract features like MFCC for automatic speech recognition.
- **Speaker Identification:** Filter banks preserve spectral envelope details that distinguish individual voices.
- **Source Separation:** Cocktail Problem: Applied in separating overlapping voices or instruments in audio recordings.
- **Speech Enhancement:** Spatial filter banks help in multichannel speech enhancement, making speech clearer in noisy environments.
- **Speech Synthesis:** Filter banks can approximate the resonances of the vocal tract (formants). Modern systems (e.g., Tacotron, WaveNet) often use Mel-spectrograms derived from filter banks as input representations.
- **Musical Analysis:** Filter banks extract timbre and harmonic features for tasks like instrument recognition, genre classification, and music transcription.

Limitations of Filter Bank Model

- The filter bank model divides the speech spectrum into several frequency bands and measures the energy in each band. However, it has some limitations:
 - **Limited frequency resolution:** Fine spectral details may be lost within each frequency band.
 - **Fixed frequency bands:** Predefined bands may not adapt to different speech characteristics.
 - **Spectral smoothing:** Important spectral peaks, such as formants, may be hidden.
 - **More parameters:** Accurate representation may require many filters.
 - **No speech-production model:** Describes spectral energy but does not directly model the vocal tract.

Linear Prediction Coding

- **Why Linear Predictive Coding (LPC)???**
 - LPC provides a useful model of the speech signal, especially for quasi-stationary voiced regions.
 - It gives a compact representation of vocal-tract characteristics that are closely related to the speech sound.
 - The analysis is mathematically tractable and relatively simple to implement in software or hardware.
 - LPC front ends have demonstrated strong performance in speech-recognition applications.
 - The model separates source/excitation behavior from the vocal-tract spectral characteristics.

Linear Prediction

- **Linear prediction** refers to the process of estimating the present speech sample as a weighted sum of its previous samples.
 - Speech samples are not independent but highly correlated i.e., If one sample has a particular value, the samples immediately before and after it are likely to have similar or related values.
 - $x'[n] = a_1x[n-1] + a_2x[n-2] + \dots + a_px[n-p]$
Where, $x'[n]$ is current speech sample; $x[n-k]$; $k = 1, 2, \dots, p$ is the previous speech samples; a_i is prediction coefficients for $i = 1, 2, \dots, p$; p is prediction order
- If the prediction is close to actual sample then there is no need to represent the entire speech waveform. Instead we can

Predicted Sample and Prediction Error

- However, the term “**linear**” doesn’t mean that the speech waveform is a straight line. Rather, it means that the previous samples are multiplied by coefficients and then added.
- The **predicted sample** can be represented as below:

$$x'[n] = \sum_{k=1}^p a_k x[n-k] + Gu(n);$$

where, G is excitation gain; $u(n)$ is excitation signal

- The **prediction error** is $e[n] = x[n] - x'[n]$
- ***For example:***
 - If $x[n-1]$ is 4 and $x[n-2]$ is 2 and the coefficients a_1 and a_2 are 0.6 and 0.2 then the predicted sample can be determined as below:

$$x'[n] = (0.6)(4) + (0.2)(2) = 2.8$$

- **Prediction Order (p):** It is the number of previous speech samples used to predict the current speech sample.
 - In simple terms, p tells that how many previous samples have to be used to predict the current sample.
 - A higher prediction order generally allows a more detailed model of the speech signal, but also increases the number of coefficients and computational complexity.
 - Practical LPC commonly uses an order related to the sampling rate and speech bandwidth.

- **LPC coefficients** are selected so that the total prediction error is minimum.
 - Error energy can be defined as $E = \sum_n e^2[n]$ with an

All Pole Interpretation of LPC

$$H(z) = G / [1 - a_1z^{-1} - a_2z^{-2} - \dots - a_pz^{-p}]$$



1. Excitation $u(n)$: The excitation is the source that produces speech. There are two basic cases:

- **Voiced speech** → periodic excitation, due to vibration of the vocal cords.
- **Unvoiced speech** → noise-like excitation, due to turbulent airflow.

So $u(n)$ represents **what is exciting the vocal tract**.

- **Gain G :** The gain controls the strength or energy of the excitation. Therefore, $Gu(n)$ means the excitation signal $u(n)$ is scaled according to its required strength.
 - For example, louder speech generally requires a larger excitation gain.
- **All pole filter $H(z)$:** The vocal tract acts approximately like a filter that modifies the excitation. LPC represents this vocal-tract filter as:

$$H(z) = \frac{G}{1 - a_1 z^{-1} - a_2 z^{-2} - a_3 z^{-3} \dots a_p z^{-p}}$$

Where, $a_1, a_2, a_3, \dots, a_p$ are the LPC coefficients. They describe the characteristics of the vocal tract.

- **Speech $s(n)$** : After the excitation passes through the vocal-tract filter, we obtain the speech signal:

$$\mathbf{u(n)} \rightarrow \mathbf{G} \rightarrow \mathbf{H(z)} \rightarrow \mathbf{s(n)}$$

- **All pole** means that the transfer function has poles but no zeros in its denominator model.
 - These poles are important because they create resonances in the vocal tract, which are related to formants.
- LPC uses the poles of an all-pole filter to represent the resonant characteristics of the vocal tract.
- LPC separates speech into two concepts: "what excites the vocal tract" and "how the vocal tract shapes that excitation."

Speech excitation in the LPC model

- **Voiced speech:** excitation is modeled as a quasi-periodic pulse train.
- **Unvoiced speech:** excitation is modeled as a random-noise sequence.
- The **voiced/unvoiced decision** controls the excitation source.
- The gain and filter coefficients vary relatively slowly with time compared with individual samples.
- Thus the LPC parameter set describes the short-time speech production mechanism.

LPC Parameter Set

- Typical model parameters are:
 - **Voiced/unvoiced classification:** vocal cords vibration creates periodic excitation (voiced); turbulent airflow creates noise like excitation (unvoiced)
 - **Pitch period for voiced speech:** Time between two consecutive vocal-cord excitation pulses.
 - **Excitation gain:** Controls the strength or energy of the excitation.
 - **LPC digital-filter coefficients:** Describe the vocal-tract filter and determine its resonant characteristics.
- Speech characteristics change with time, but they change relatively slowly compared with individual samples. Therefore, speech is divided into short frames, typically around 20–30 ms, and these parameters are estimated for each frame.

Autocorrelation Method

- The **Autocorrelation** method is a technique used in LPC to determine the prediction coefficients by measuring the similarity between a speech signal and its delayed version.
- The autocorrelation of a speech signal is given by:

$$R[k] = \sum_n x[n]x[n - k]$$

where, $R[k]$ is autocorrelation at lag k ; $x[n]$ is speech sample; $x[n-k]$ is delayed speech sample; k is lag

- The autocorrelation method is computationally attractive because the matrix structure can be exploited efficiently.
- The analysis minimizes the mean-squared prediction error over the selected frame.

- Autocorrelation Method:
 - Window the speech frame and compute the autocorrelation sequence $R(k)$.
 - The zeroth autocorrelation $R(0)$ represents the frame energy.
 - Use $R(1) \dots R(p)$ to form the LPC normal equations.
 - Solve for the predictor coefficients $a_1 \dots a_p$ using

$$\begin{bmatrix} R(0) & R(1) & \cdots & R(p-1) \\ R(1) & R(0) & \cdots & R(p-2) \\ \vdots & \vdots & \ddots & \vdots \\ R(p-1) & R(p-2) & \cdots & R(0) \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix} = \begin{bmatrix} R(1) \\ R(2) \\ \vdots \\ R(p) \end{bmatrix}$$

- The method avoids explicit handling of samples outside the analysis frame by the windowing operation.

Covariance Method

- The covariance method uses the actual samples within the analysis interval to construct the prediction-error equations.
- It does not force the signal outside the selected interval to zero through an autocorrelation-style extension.
- The resulting equations are based on sample products over the valid prediction region.
- Both covariance and autocorrelation approaches estimate an LPC model by minimizing prediction error; their boundary handling differs.

Autocorrelation vs Covariance

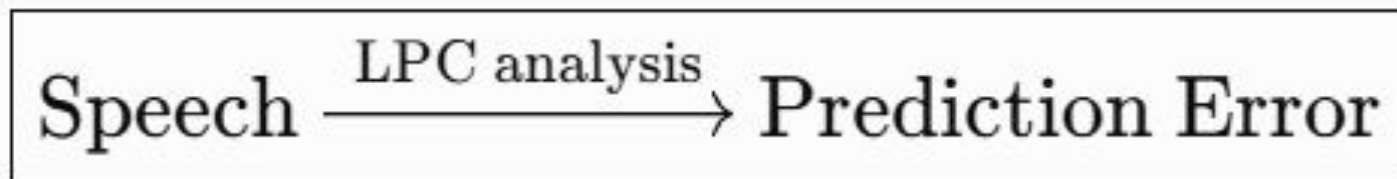
Aspect	Autocorrelation	Covariance
Boundary treatment	Uses a windowed sequence and its autocorrelation	Uses samples directly over the valid prediction region
Matrix structure	Symmetric Toeplitz structure	General covariance structure
Typical use	Widely used practical LPC method	Alternative formulation for LPC analysis
Key idea	Minimize energy of the windowed prediction error	Minimize prediction error using available samples

Examples of LPC analysis

- We can compare **speech waveforms, prediction errors, FFT spectra and LPC spectra** for representative speech segments.
 - **Speech waveform** shows the actual speech signal in the time domain. It contains both the vocal-tract characteristics and the excitation/source information.
 - **Prediction-error signal:** LPC tries to predict the current speech sample from previous samples. The difference between the actual and predicted sample is the prediction error.
 - **FFT spectrum** shows the detailed frequency content of the speech segment, including many individual harmonics for voiced speech.

–**LPC spectrum** gives a smooth spectral envelope rather than showing every individual harmonic. The prominent peaks, called formants, represent the major resonances of the vocal tract.

- These examples illustrate the '**whitening**' behavior of LPC analysis. The idea of **whitening** is that LPC filtering removes the predictable spectral structure of speech.



- The prediction error therefore has a flatter spectrum than the original speech, ideally approaching a white/noise-like spectrum.

LPC front-end processor for Speech Recognition

- **Preemphasis:** Pre-emphasis compensates for the natural spectral tilt of speech by boosting higher-frequency components, typically using a first-order filter.
 - It conditions the signal before dividing it into short, approximately stationary frames for LPC analysis.
- **Frame blocking** segments the speech waveform into short overlapping frames so that the signal can be treated as approximately stationary.
- **Autocorrelation** calculates $p+1$ values for each speech frame, where the zeroth value represents the signal energy and the higher-lag values show similarity between samples. These values are then used to calculate the LPC coefficients.

- **LPC Analysis:** The LPC coefficients are calculated by finding a model that predicts the current speech sample from previous samples. The LPC model mainly represents the vocal-tract characteristics of speech.
- **LPC Parameter Conversion:** The LPC coefficients can be converted into other representations such as cepstral coefficients or reflection coefficients, which may be more suitable for recognition.
- **Parameter Weighting:** Different LPC-derived parameters are weighted to improve their usefulness and reduce the effect of unwanted variations, such as noise or sensitivity to certain coefficients.
- **Temporal Cepstral Derivative:** The change in cepstral features over time is calculated, often called delta (Δ) features. These capture the dynamic information of speech.

- The front end produces a sequence of **feature vectors**, one vector per analysis frame.
- Each vector can contain weighted LPC-derived cepstral coefficients, parameter weighting, and temporal cepstral derivatives.
- Thus, the speech signal becomes a sequence of feature vectors: $F_1, F_2, F_3, \dots, F_n$
- This sequence is finally given to the speech-recognition system, which compares the features with stored speech patterns/models to determine what was spoken.
- LPC front-end processing converts speech into a sequence of compact feature vectors that describe both the spectral characteristics and temporal changes of speech for recognition.

Characteristics of LPC

- LPC models short-time speech using a linear predictor and an all-pole vocal-tract model.
- Autocorrelation-based LPC converts frame statistics into predictor parameters efficiently.
- The LPC front end consists of signal conditioning, framing, autocorrelation, LPC analysis and feature conversion.
- LPC-derived cepstral coefficients provide a compact spectral representation for recognition.
- Parameter weighting and temporal derivatives improve the representation by reducing sensitivity and adding dynamics.
- Practical systems choose frame size, frame shift, LPC order and feature dimension together.